GREEN VALLEY DIAGNOSTIC CENTRE

SURGICAL PATHOLOGY & HISTOPATHOLOGY REPORT

Patient ID:	SYN-CANCER-001 (Synthetic)	**Document ID:**	doc-002
Document Date:	2026-01-08	**Document Type:**	Pathology Report
Facility:	Green Valley Diagnostic Centre	**Department:**	Department of Anatomical Pathology

SPECIMEN IDENTIFICATION & CLINICAL DATA

Specimen: Core Needle Biopsy, Right Breast.

Procedure Date: 2026-01-08

Clinical History: Palpable right breast mass in upper-outer quadrant.

GROSS / MACROSCOPIC DESCRIPTION

Received in formalin labeled with synthetic patient ID SYN-CANCER-001 are four (4) core needle biopsy fragments. The tissue cores range in length from 1.1 cm to 1.6 cm, with a uniform diameter of 0.1 cm. Color is tan-white to fibrofatty. Entire specimen submitted in one cassette (Block A1).

PROCESSING PROTOCOL

Standard formalin fixation (18 hours), automated paraffin embedding, 4-micron microtomy sections, and routine Hematoxylin & Eosin (H&E;) staining performed. Immunohistochemical (IHC) profiling initiated.

MICROSCOPIC DESCRIPTION & FINAL DOCUMENTED FINDINGS

Sections demonstrate invasive carcinoma with glandular and corded architectural patterns. Tubule formation is moderate. Nuclei exhibit moderate pleomorphism with occasional visible mitotic figures. Surrounding stroma reveals desmoplastic reaction with minimal lymphocytic infiltrate.

Documented Histopathology Diagnosis:
Pathology report documents: Invasive ductal carcinoma, histological Nottingham grade 2. Estrogen receptor (ER) positive (85%), Progesterone receptor (PR) positive (70%), HER2 negative (score 1+ by IHC). Ki-67 index: 15%.

IMMUNOHISTOCHEMICAL BIOMARKER PROFILE

Biomarker	Method	Documented Result	Reference Criteria
Estrogen Receptor (ER)	IHC (Clone SP1)	Positive (85% strong nuclear staining)	ASCO/CAP >= 1% positive
Progesterone Receptor (PR)	IHC (Clone 1E2)	Positive (70% moderate nuclear staining)	ASCO/CAP >= 1% positive
HER2 / neu Protein	IHC (Clone 4B5)	Negative (Score 1+ faint membrane staining)	ASCO/CAP Guidelines
Ki-67 Proliferation Index	IHC (Clone MIB-1)	15% nuclear labeling index	Documented count per 1000 cells

PATHOLOGIST SIGN-OFF

Reporting Pathologist: Green Valley Diagnostic Pathology Staff